ModelIt!

Student Activity Pack

The Reef Is Bleaching

Why the World's Most Colorful Ecosystems Are Turning White

Grade 8 | NGSS MS-LS2-1, MS-LS2-4 | Stability and Change

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. What do you know about coral reefs? Are they alive?

2. What do you think causes coral bleaching?

3. How might rising ocean temperatures affect marine life?

4. Draw a food web that might exist on a coral reef.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Ocean Temperature | Ocean Acidity (pH) | Zooxanthellae Population | Coral Health

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? **Explain ONE relationship in your model. What happens when one component changes?**

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Healthy Reef

Normal ocean temperature and pH

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heat Wave Simulation

Lock Ocean Temperature to +3°C above normal

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Double Threat

Increase both temperature and acidity

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Fish Population — Fish depend on coral for food and shelter

Algae Overgrowth — Without coral, algae can take over the reef

Human Tourism — Tourism dollars fund conservation but also cause damage

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are coral reefs turning white and dying? Use evidence from your simulation to support your answers.

1. **Why is it called 'bleaching' if the coral isn't being washed with bleach?**

2. **How is the coral-zooxanthellae relationship similar to a business partnership?**

3. **If coral reefs disappear, what happens to the 25% of marine species that depend on them?**

4. **Why is the combination of heat and acidity worse than either factor alone?**

5. **What can an 8th grader actually do to help coral reefs survive?**

8 STEM Challenge: Design a Coral Reef Rescue Plan

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YOUR ENGINEERING MISSION

Design a science-based plan to protect and restore a bleaching coral reef using evidence from your model.

THE SCENARIO

The Great Barrier Reef has experienced its worst bleaching event ever. Your marine science team has been hired to propose a rescue and prevention plan.

Engineering Design Thinking

1. WHAT IMMEDIATE ACTIONS CAN PROTECT EXISTING CORAL?
2. HOW WOULD YOU RESTORE ZOOXANTHELLAE POPULATIONS?
3. WHAT LONG-TERM CLIMATE ACTIONS WOULD PREVENT FUTURE BLEACHING?
4. HOW WILL YOU MEASURE THE SUCCESS OF YOUR PLAN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Marine Biologists and Coral Restoration Scientists work to understand and protect ocean ecosystems. They earn $60,000-$100,000/year and work in some of the world's most beautiful locations.